

Feature Engineering Report

Project Title

Enterprise Data Preparation for Employee Attrition

Introduction

Feature engineering is the process of creating new and meaningful variables from existing data. The purpose is to represent employee information in a way that can provide deeper insights into factors related to employee attrition.

In this project, several derived HR features were created using employee income, work experience, company tenure, promotion history, and relationship with the current manager.

1. Income Per Year of Experience

This feature was created by dividing MonthlyIncome by TotalWorkingYears plus one.

Formula:

$\text{MonthlyIncome} / (\text{TotalWorkingYears} + 1)$

This feature provides an approximate relationship between an employee's income and overall work experience. It may help identify whether compensation is relatively low or high compared with experience.

2. Promotion Delay Ratio

This feature was created using YearsSinceLastPromotion and YearsAtCompany.

Formula:

$\text{YearsSinceLastPromotion} / (\text{YearsAtCompany} + 1)$

This feature represents how long an employee has waited for a promotion compared with their time in the organization. A high value may indicate that an employee has spent a significant part of their company tenure without a recent promotion.

3. Career Stability Ratio

This feature compares YearsAtCompany with TotalWorkingYears.

Formula:

$\text{YearsAtCompany} / (\text{TotalWorkingYears} + 1)$

It provides an indication of how much of an employee's overall career has been spent in the current organization. A higher value may represent stronger long-term association with the company.

4. Manager Stability Ratio

This feature was created by comparing YearsWithCurrManager with YearsAtCompany.

Formula:

$\text{YearsWithCurrManager} / (\text{YearsAtCompany} + 1)$

This ratio gives information about the stability of the employee-manager relationship. Frequent manager changes may influence employee satisfaction and workplace experience.

5. Income Per Job Level

This feature combines MonthlyIncome and JobLevel.

Formula:

$\text{MonthlyIncome} / (\text{JobLevel} + 1)$

This feature helps examine the relationship between salary and job level. It may provide additional insight into employee compensation patterns.

Importance of Feature Engineering

The original HR dataset contains many useful individual variables. However, relationships between features can sometimes provide more meaningful information than the original variables alone.

The derived features created in this project combine salary, experience, tenure, promotions, job level, and management relationships. These features may help future machine learning models identify more complex patterns related to employee attrition.

Conclusion

A total of five derived HR features were created during the feature engineering process. These features provide additional information about employee compensation, career growth, company stability, promotion history, and management stability.

The engineered features improve the analytical value of the dataset and can be used in future employee attrition prediction models.